ZOMBIE RUNNER SUBWAY TUNNEL

3D Survival Horror Endless Runner | Unity Project Report

Course: Game Mechanics, Design and Development **Course Code:** 23CS201PE407

1. Introduction

ZOMBIE RUNNER SUBWAY TUNNEL is a 3D survival horror endless runner made in Unity 6. The player takes the role of an armed survivor running through abandoned city streets and subway tunnels. The game combines three-lane movement and obstacle avoidance with zombie pursuit, firearm combat, collectible resources, and horror atmosphere.

The player shifts between the left, centre, and right lanes, jumps over barricades, slides beneath low barriers, and shoots threats ahead. As the run continues, forward speed rises and the route streams between city and underground subway environments.

Game objective: Survive for as long as possible, avoid hazards, defeat pursuing zombies, collect coins and supplies, and build a high score.

2. Game Design and Mechanics

Game element	Implemented design and mechanics
Three-lane running	Left, centre, and right lanes are positioned at X = -3 m, 0 m, and +3 m. Forward speed starts at 8 m/s and increases to a 16 m/s maximum.
Jump and slide	Grounded jump clears barricades. Jump height is 1.8 m. A 0.8-second slide lowers the player collision height to pass beneath tunnel beams and barriers.
Zombie pursuit	Three configurable zombie types pursue and attack: Common Runner, Fast Sprinter, and Heavy Brute. Their animation states cover idle, walk, run, attack, hit, and death.
Firearm combat	Forward raycast shooting uses auto-aim within a 12-degree cone and 32 m range. Bullet tracers, recoil, muzzle flash, and pooled hit effects show shot feedback.
Endless environment	City and subway routes use five reusable 30 m segments. Biomes change every 300 m; the pool recycles segments and avoids repeated runtime instantiation.
Run resources	Coins, ammunition, and medical kits support the run. The HUD tracks distance, score, coins, ammunition, health, and active power-up time.
Audio and horror mood	Four adaptive music layers move between exploration, tension, combat, and critical health. Spatial zombie sounds help signal nearby threats.

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2. Game Design and Mechanics continued

Controls

Action	Keyboard	Secondary input
Move left / right	A / D	Left / right arrow or touch swipe
Jump / vault	W or Space	Up arrow or swipe up
Slide / duck	S	Down arrow or swipe down
Fire weapon	F	Left mouse button or on-screen fire
Pause / resume	Escape	P or pause button

Zombie archetypes

Archetype	Health	Speed	Damage / range	Points
Common Runner	3 HP	5.0 m/s	10 / 1.8 m	100
Fast Sprinter	2 HP	7.5 m/s	8 / 1.6 m	150
Heavy Brute	8 HP	3.2 m/s	25 / 2.2 m	250

Collectibles and power-ups

Item	Effect in the game
Ammunition box	Adds 15 rounds.
Medical kit	Restores 25 HP.
Energy Shield	Absorbs one lethal collision or attack.
Coin Magnet	Draws coins from within an 8 m radius.
Speed Boost	Adds 50% forward speed and temporary invulnerability.
Double Coins	Doubles coin rewards while active.

Run score: Distance + kills. Coins collected during a run are banked when the run ends. High scores and banked coins persist between sessions.

3. Step-by-Step Game Development Procedure

Step 1: Configure the Unity project

The game was developed in Unity 6 (6000.5.9f1) with Universal Render Pipeline 17.5.0 and C# on .NET Standard 2.1. The Unity Input System handles keyboard, mouse, and touch input. TextMeshPro and uGUI provide the interface.

Step 2: Implement the survivor controls

The player uses a CharacterController and three lane positions at -3 m, 0 m, and +3 m. Lane changes interpolate toward the selected lane. Forward speed rises from 8 m/s toward 16 m/s. Grounded jumping reaches 1.8 m; sliding lasts 0.8 seconds and adjusts the collision capsule without deforming the humanoid model.

Step 3: Build the city and subway route

The environment streams five pooled 30 m segments, keeping 150 m of route active. When a segment passes behind the player, it moves to the front of the queue and receives fresh obstacle and collectible placement. The visual theme switches between abandoned city ruins and a detailed subway tunnel every 300 m. Fog, blue moonlight, street lamps, tunnel fixtures, and a weapon-mounted flashlight set the night-time atmosphere.

Step 4: Add zombie behaviour and combat

Zombie archetype statistics are stored as ScriptableObject data. A Mecanim state machine drives each enemy through movement, attack, hit reaction, and death. The weapon system checks targets with a forward raycast and auto-aim cone, then applies damage and plays tracer, recoil, light-flash, and impact effects. Reusable particle pools reduce runtime allocation.

Step 5: Connect pickups, interface, and run flow

Coins, ammunition, health kits, and four timed power-ups connect to score and survival systems. The HUD shows distance, score, coins, ammunition, health, and power-up timers. At defeat, the game displays run results and saved high score. A single shielded revive restores full health, 30 rounds, and three seconds of invulnerability.

Step 6: Test and prepare builds

The project was checked for lane bounds, jump and slide timing, combat hit registration, zombie defeat, power-up timers, and object-pool recycling. The report records a verified Windows PC build and an Android APK build configured for ARM64. Profiling targeted stable 60 FPS operation with minimal garbage-collection pauses.

4. Game Screenshots

The screenshots show the implemented menu and the in-game city route, including the survivor, zombie targets, weapon reticle, and run HUD.



Figure 1. Main menu and 3D survivor staging.



Figure 2. In-game city route and survival HUD.

5. Expected Game Output

A completed run presents a continuously changing route through city streets and subway tunnels. The player reacts to barriers and approaching zombies by shifting lanes, jumping, sliding, or firing. Speed increases with distance while coins, supplies, score, and enemy encounters keep the run active. The game-over screen shows run statistics and record status, with retry, menu, and one-time revive options.

On supported mobile layouts, touch input and safe-area positioning keep movement controls and status information visible. The settings menu provides graphics, audio, and control guides.

6. Conclusion

ZOMBIE RUNNER SUBWAY TUNNEL combines endless running with survival horror combat. Its three-lane controls, escalating speed, distinct zombie archetypes, pooled city and subway segments, pickups, adaptive audio, and persistent run records form a cohesive gameplay loop. The project also demonstrates modular Unity development through separate player, enemy, environment, combat, audio, and interface systems.